The RSOPT Project

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Quadratic Programs

This chapter covers various forms of **quadratic programs** supported by the RSOPT project. This document defines quadratic programs as convex programs whose objective has a constant Hessian with no negative eigenvalues.

3.1 Unconstrained Quadratic Programs

Unconstrained quadratic programs (`UnconstrainedQuadraticProgram`) are convex programs of the form

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top A x + b^\top x \tag{3.1}$$

where $A \in \mathbb{R}^{n \times n}$, $A \succeq 0$, and $b \in \mathbb{R}^n$. The relevant stationarity condition is

$$Ax^\star + b = 0 \Leftrightarrow x^\star = -A^{-1}b. \tag{3.2}$$

We solve unconstrained quadratic programs using the **conjugate gradient** method. TODO